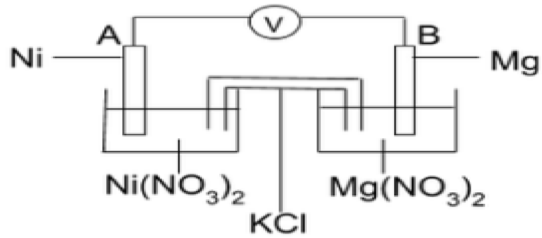
In the cell below a nickel electrode is connected to a magnesium electrode. The cell is set up under standard conditions. Both electrodes are placed in electrolytes, connected with a salt bridge.



1.1 Write down the energy conversion that occurs in this cell. (2)

1.2 Define the term **electrolyte**. (2)

1.3 Which electrode, **A** or **B**, is the anode? (1)

1.4 Write down the reduction half-reaction that occurs in this cell. (2)

1.5 Calculate the reading on the voltmeter for this cell. (4)

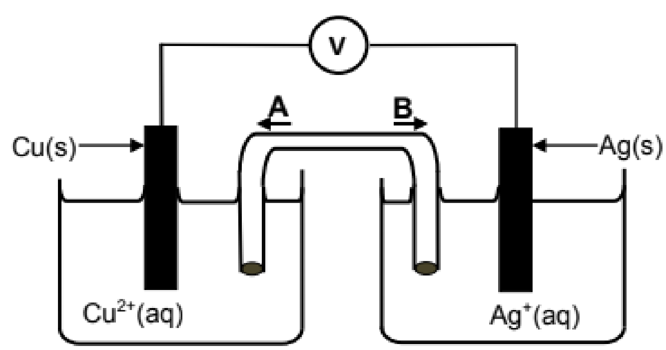
The voltmeter is now replaced with an ammeter.

1.6 In which direction do the electrons flow in the external circuit? Write down ONLY from **A to B** or from **B to A**? (1)

1.7 Explain the function of the salt bridge in this cell by referring to the movement of electrons and ions. (3)

1.8 Write down the cell notation for this cell. (3)

The learner now sets up a galvanic cell as shown below. The cell functions under standard conditions.



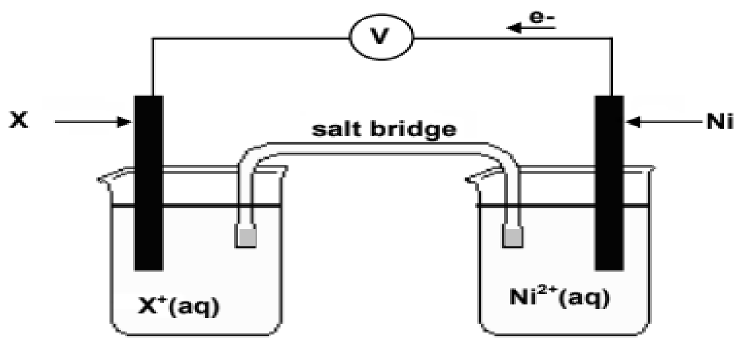
2.1 Write down the energy conversion that takes place in this cell. (1)

2.2 In which direction (**A** or **B**) will ANIONS move in the salt bridge? (1)

2.3 Calculate the emf of the above cell under standard conditions. (4)

2.4 Write down the balanced equation for the net cell reaction that takes place in this cell. (3)

An electrochemical cell is set up at **standard conditions** as shown below. The electron flow is as indicated. The reading on the voltmeter is **1,07 V**.



3.1 Write down ONE function of the salt bridge. (1)

3.2 Which electrode of the cell is the anode? Write only **X** or **Ni**. (1)

3.3 Using a calculation, identify the unknown metal **X**. (5)

3.4 Write down the balanced net (overall) equation for the above cell. (3)

Metal **X** is now replaced by zinc.

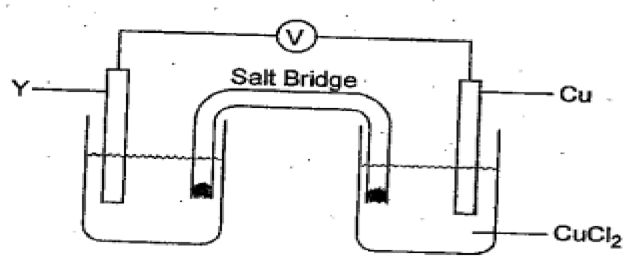
3.5 For the new galvanic cell, write down:

3.5.1 The NAME or FORMULA of the oxidising agent (1)

3.5.2 The half reaction for the ANODE (2)

3.5.3 The cell notation of the new cell (3)

An electrochemical cell is setup under standard conditions as shown below. Copper (Cu) is used as the cathode and an unknown metal Y as the anode of the cell. The voltmeter connected across the two electrodes shows an initial reading of 2,70 V.



4.1 State the standard conditions that are applicable to this cell. (2)

4.2 State the energy conversion that takes place in this cell. (1)

4.3 State two functions of the salt bridge. (2)

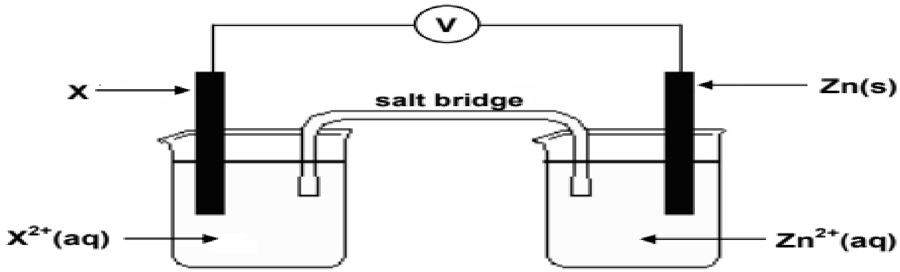
4.4 Use the information given to identify the metal Y. Show all calculations. (5)

4.5 Write down the oxidation half reaction. (2)

4.6 Write down the cell notation (symbolic notation) of this cell. (3)

The cell notation of a standard galvanic cell that contains an unkown metal electrode **X**, is as follows:

$$Zn(s) \mid Zn^{2+}(1\ mol \cdot dm^{-3}) \parallel \mathbf{X}^{2+}(1\ mol \cdot dm^{-3}) \mid \mathbf{X}(s)$$



5.1 Name the component of the cell that is represented by the double vertical lines (||) in the cell notation. (1)

5.2 Name TWO standard conditions that is applicable to the $Zn^{2+}|Zn$-half cell. (2)

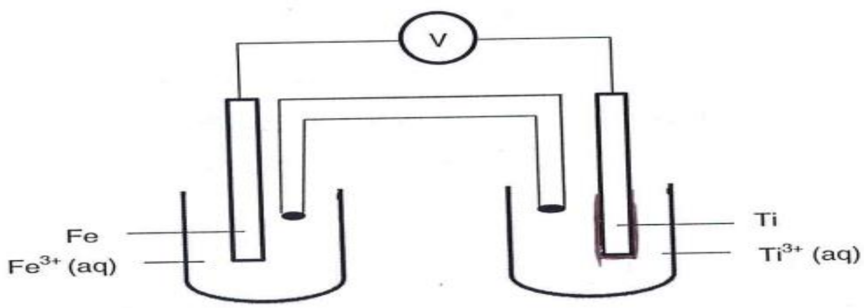
5.3 Identify the reducing agent in the above mentioned cell. (1)

5.4 The initial reading on the volt meter connected to the electrodes of the above given cell is 0,49 V. Identify metal **X** by calculating the standard reduction potential of the unkown metal **X**. (5)

5.5 Write a balanced chemical equation for the net reaction that takes place in this cell. Leave out the spectator ions. (3)

A galvanic cell is set up based on the reaction represented by the equation below. The reaction takes place under standard conditions.

$$Fe^{3+} + Ti \rightleftharpoons Fe + Ti^{3+}$$



6.1 Define the term *oxidation* in terms of oxidation number. (2)

6.2 State TWO functions of the salt bridge. (2)

6.3 Write down the:

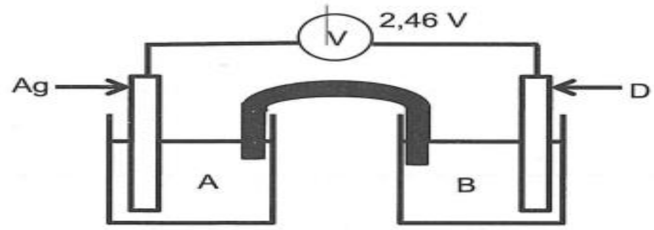
6.3.1 Half-reaction that takes place at the cathode. (2)

6.3.2 Cell notation for this cell. (3)

6.4 The initial reading on the voltmeter is 1,57 V. Calculate the standard reduction potential of the Ti electrode. (4)

6.5 Write down the value of the reading on the voltmeter when the cell reaction reaches equilibrium. (1)

A standard electrochemical cell is set up, using silver and an unknown electrode, D, as shown in the sketch below,. The initial reading on the voltmeter is 2,46 V. After the cell was connected for some time, the mass of the silver electrode increased.



8.1 What process takes place at the silver electrode? Choose from OXIDATION or REDUCTION. (1)

8.2 Define an *oxidising agent* in terms of electron transfer. (2)

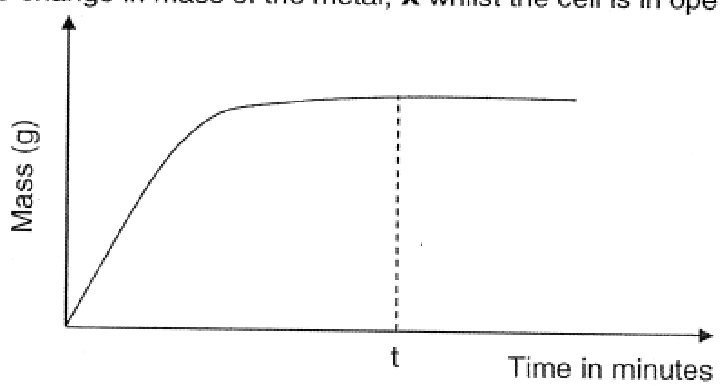
8.3 Write down the NAME of a solution which can be used in beaker A. (1)

8.4 Do electrons move in the external circuit from D to the silver electrode or from the silver electrode to D? (1)

8.5 Determine the standard electrode potential for electrode D and identify D from the standard potential table. (5)

8.6 Write down the cell notation for this cell. (3)

A standard galvanic cell is set up using a zinc and an unknown metal, **X**. The graph below shows the change in mass of the metal, **X** whilst the cell is in operation.



8.1 Define an electrolyte. (2)

8.2 Write down the half reaction that takes place at the anode of this cell. (2)

8.3 The initial emf of this cell under standard conditions is 0,63 V.

8.3.1 Write down TWO standard conditions required for the initial emf to be 0,63 V (2)

8.3.2 Identify the metal **X**, by calculating the standard reduction potential of **X**. (5)

8.4 Write down the cell notation for this cell. (3)

8.5 Write down the value of the emf at time **t**, shown on the graph. (1)